

# CS 506: Introduction to Quantum Computing

Lecture Notes: Quantum Teleportation, Approximate Computation,  
Phase Kickback, and the Deutsch Problem

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## 1 Quantum Teleportation

### 1.1 Setup

Alice wants to send a qubit

$$|\psi\rangle = \alpha_0 |0\rangle + \alpha_1 |1\rangle, \quad |\alpha_0|^2 + |\alpha_1|^2 = 1,$$

to Bob via a **classical channel**. The plan:

- Alice will send **two classical bits** to Bob.
- Bob will figure out  $|\psi\rangle$ .

### 1.2 Alice and Bob share a Bell pair

Beforehand, Alice and Bob meet and create a 2-qubit system:

$$|\beta_{00}\rangle = \frac{1}{\sqrt{2}} |0\rangle |0\rangle + \frac{1}{\sqrt{2}} |1\rangle |1\rangle.$$

**Alice takes the first qubit and Bob takes the second.** Later, Alice gets  $|\psi\rangle$  and wants to send it to Bob.

The Bell states used below are

$$\begin{aligned} |\beta_{00}\rangle &= \frac{1}{\sqrt{2}} |00\rangle + \frac{1}{\sqrt{2}} |11\rangle, & |\beta_{01}\rangle &= \frac{1}{\sqrt{2}} |01\rangle + \frac{1}{\sqrt{2}} |10\rangle, \\ |\beta_{10}\rangle &= \frac{1}{\sqrt{2}} |00\rangle - \frac{1}{\sqrt{2}} |11\rangle, & |\beta_{11}\rangle &= \frac{1}{\sqrt{2}} |01\rangle - \frac{1}{\sqrt{2}} |10\rangle. \end{aligned}$$

### 1.3 The total 3-qubit system

$$|\psi\rangle \otimes |\beta_{00}\rangle = |\psi\rangle \otimes \left( \frac{1}{\sqrt{2}} |0\rangle \otimes |0\rangle + \frac{1}{\sqrt{2}} |1\rangle \otimes |1\rangle \right).$$

Alice holds the first two qubits ( $|\psi\rangle$  and her half of  $|\beta_{00}\rangle$ ). Bob holds the third.

### Claim (prove it!)

$$|\psi\rangle \otimes |\beta_{00}\rangle = \frac{1}{2} |\beta_{00}\rangle \otimes |\psi\rangle + \frac{1}{2} |\beta_{01}\rangle \otimes (X|\psi\rangle) + \frac{1}{2} |\beta_{10}\rangle \otimes (Z|\psi\rangle) + \frac{1}{2} |\beta_{11}\rangle \otimes (XZ|\psi\rangle).$$

Each Bell state is on Alice's two qubits. The state next to it is Bob's qubit.

## 1.4 Proof: expand both sides and show they are the same

The Pauli-transformed states on the right side are

$$X|\psi\rangle = \alpha_0|1\rangle + \alpha_1|0\rangle, \quad Z|\psi\rangle = \alpha_0|0\rangle - \alpha_1|1\rangle, \quad XZ|\psi\rangle = \alpha_0|1\rangle - \alpha_1|0\rangle.$$

**Left side.**

$$\begin{aligned} \text{LHS} &= (\alpha_0|0\rangle + \alpha_1|1\rangle) \left( \frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|11\rangle \right) \\ &= \frac{1}{\sqrt{2}}\alpha_0|000\rangle + \frac{1}{\sqrt{2}}\alpha_0|011\rangle + \frac{1}{\sqrt{2}}\alpha_1|100\rangle + \frac{1}{\sqrt{2}}\alpha_1|111\rangle. \end{aligned}$$

**Right side.**

$$\begin{aligned} \text{RHS} &= \frac{1}{2} \left( \frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|11\rangle \right) (\alpha_0|0\rangle + \alpha_1|1\rangle) \\ &\quad + \frac{1}{2} \left( \frac{1}{\sqrt{2}}|01\rangle + \frac{1}{\sqrt{2}}|10\rangle \right) (\alpha_0|1\rangle + \alpha_1|0\rangle) \\ &\quad + \frac{1}{2} \left( \frac{1}{\sqrt{2}}|00\rangle - \frac{1}{\sqrt{2}}|11\rangle \right) (\alpha_0|0\rangle - \alpha_1|1\rangle) \\ &\quad + \frac{1}{2} \left( \frac{1}{\sqrt{2}}|01\rangle - \frac{1}{\sqrt{2}}|10\rangle \right) (\alpha_0|1\rangle - \alpha_1|0\rangle). \end{aligned}$$

Factor out  $\frac{1}{2\sqrt{2}}$  and expand each product:

|                     |                                                                                         |
|---------------------|-----------------------------------------------------------------------------------------|
| $\beta_{00}$ term : | $\alpha_0 000\rangle + \alpha_1 001\rangle + \alpha_0 110\rangle + \alpha_1 111\rangle$ |
| $\beta_{01}$ term : | $\alpha_0 011\rangle + \alpha_1 010\rangle + \alpha_0 101\rangle + \alpha_1 100\rangle$ |
| $\beta_{10}$ term : | $\alpha_0 000\rangle - \alpha_1 001\rangle - \alpha_0 110\rangle + \alpha_1 111\rangle$ |
| $\beta_{11}$ term : | $\alpha_0 011\rangle - \alpha_1 010\rangle - \alpha_0 101\rangle + \alpha_1 100\rangle$ |

Collect coefficients:

| Basis state   | Sum of coefficients               | Times $\frac{1}{2\sqrt{2}}$ |
|---------------|-----------------------------------|-----------------------------|
| $ 000\rangle$ | $\alpha_0 + \alpha_0 = 2\alpha_0$ | $\alpha_0/\sqrt{2}$         |
| $ 001\rangle$ | $\alpha_1 - \alpha_1 = 0$         | 0                           |
| $ 010\rangle$ | $\alpha_1 - \alpha_1 = 0$         | 0                           |
| $ 011\rangle$ | $\alpha_0 + \alpha_0 = 2\alpha_0$ | $\alpha_0/\sqrt{2}$         |
| $ 100\rangle$ | $\alpha_1 + \alpha_1 = 2\alpha_1$ | $\alpha_1/\sqrt{2}$         |
| $ 101\rangle$ | $\alpha_0 - \alpha_0 = 0$         | 0                           |
| $ 110\rangle$ | $\alpha_0 - \alpha_0 = 0$         | 0                           |
| $ 111\rangle$ | $\alpha_1 + \alpha_1 = 2\alpha_1$ | $\alpha_1/\sqrt{2}$         |

So  $\text{RHS} = \frac{1}{\sqrt{2}}\alpha_0|000\rangle + \frac{1}{\sqrt{2}}\alpha_0|011\rangle + \frac{1}{\sqrt{2}}\alpha_1|100\rangle + \frac{1}{\sqrt{2}}\alpha_1|111\rangle = \text{LHS}$ . ■

## 1.5 The protocol

Alice measures her first two qubits in the Bell basis. Each outcome has probability  $\left(\frac{1}{2}\right)^2 = \frac{1}{4}$ . She sends the matching two bits, and Bob fixes his qubit:

| Alice measures<br>(Bell basis) | Alice<br>sends | Bob's modified<br>qubit | Bob does               | Check                                              |
|--------------------------------|----------------|-------------------------|------------------------|----------------------------------------------------|
| $ \beta_{00}\rangle$           | 00             | $ \psi\rangle$          | nothing, applies $I$   | $I \psi\rangle =  \psi\rangle$                     |
| $ \beta_{01}\rangle$           | 01             | $X \psi\rangle$         | applies $X$            | $X(X \psi\rangle) =  \psi\rangle$                  |
| $ \beta_{10}\rangle$           | 10             | $Z \psi\rangle$         | applies $Z$            | $Z(Z \psi\rangle) =  \psi\rangle$                  |
| $ \beta_{11}\rangle$           | 11             | $XZ \psi\rangle$        | applies $X$ , then $Z$ | $ZXXZ \psi\rangle = ZZ \psi\rangle =  \psi\rangle$ |

In the last row,  $XX = I$ , which leaves  $ZZ = I$ .

## 2 Approximate Computation

### Definition

Given a unitary operator  $U$ , we design a circuit that computes a unitary  $V$ . For  $\varepsilon > 0$ , “ $V$  approximates  $U$  within error  $\varepsilon$ ” means

$$\max_{|\psi\rangle} \|V|\psi\rangle - U|\psi\rangle\| \leq \varepsilon.$$

## 3 Phase Kickback

### 3.1 CNOT with target $\frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$

The first qubit is the control and the second is the target.

$$\begin{aligned} \text{CNOT}\left(|0\rangle\left(\frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle\right)\right) &= \text{CNOT}\left(|0\rangle\frac{1}{\sqrt{2}}|0\rangle\right) + \text{CNOT}\left(|0\rangle\frac{1}{\sqrt{2}}|1\rangle\right) \\ &= \frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|01\rangle = |0\rangle\left(\frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle\right), \\ \text{CNOT}\left(|1\rangle\left(\frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle\right)\right) &= \text{CNOT}\left(|1\rangle\frac{1}{\sqrt{2}}|0\rangle\right) + \text{CNOT}\left(|1\rangle\frac{1}{\sqrt{2}}|1\rangle\right) \\ &= \frac{1}{\sqrt{2}}|11\rangle + \frac{1}{\sqrt{2}}|10\rangle = |1\rangle\left(\frac{1}{\sqrt{2}}|1\rangle + \frac{1}{\sqrt{2}}|0\rangle\right). \end{aligned}$$

### 3.2 CNOT with target $\frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}|1\rangle$

$$\begin{aligned}\text{CNOT}\left(|0\rangle\left(\frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}|1\rangle\right)\right) &= \frac{1}{\sqrt{2}}\text{CNOT}(|0\rangle|0\rangle) - \frac{1}{\sqrt{2}}\text{CNOT}(|0\rangle|1\rangle) \\ &= \frac{1}{\sqrt{2}}|0\rangle|0\rangle - \frac{1}{\sqrt{2}}|0\rangle|1\rangle = +1 \cdot |0\rangle\left(\frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}|1\rangle\right), \\ \text{CNOT}\left(|1\rangle\left(\frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}|1\rangle\right)\right) &= \frac{1}{\sqrt{2}}\text{CNOT}(|1\rangle|0\rangle) - \frac{1}{\sqrt{2}}\text{CNOT}(|1\rangle|1\rangle) \\ &= \frac{1}{\sqrt{2}}|1\rangle|1\rangle - \frac{1}{\sqrt{2}}|1\rangle|0\rangle = -1 \cdot |1\rangle\left(\frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}|1\rangle\right).\end{aligned}$$

#### Result

For  $b \in \{0, 1\}$ :

$$\text{CNOT}\left(|b\rangle\left(\frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}|1\rangle\right)\right) = (-1)^b |b\rangle\left(\frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}|1\rangle\right).$$

- Eigenvalue:  $(-1)^b$ .
- Eigenvector:  $|b\rangle\left(\frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}|1\rangle\right)$ , where  $\frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}|1\rangle$  is an eigenvector (eigenstate) for the 2nd qubit.

### 3.3 Control in superposition

$$\begin{aligned}\text{CNOT}\left((\alpha_0|0\rangle + \alpha_1|1\rangle)\left(\frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}|1\rangle\right)\right) &= \alpha_0|0\rangle\left(\frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}|1\rangle\right) - \alpha_1|1\rangle\left(\frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}|1\rangle\right) \\ &= (\alpha_0|0\rangle - \alpha_1|1\rangle)\left(\frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}|1\rangle\right).\end{aligned}$$

The target qubit is unchanged. The sign is *kicked back* onto the control:  $\alpha_0|0\rangle + \alpha_1|1\rangle$  becomes  $\alpha_0|0\rangle - \alpha_1|1\rangle$ .

## 4 The Deutsch Problem

#### Deutsch problem

Given a black box computing

$$f : \{0, 1\} \rightarrow \{0, 1\},$$

we want to determine whether

$$f(0) = f(1) \quad (\text{constant}) \quad \text{or} \quad f(0) \neq f(1) \quad (\text{balanced}).$$